

WORKFORCE MANAGEMENT SYSTEM

Product Requirements Document · React + TailwindCSS + Supabase + PostgreSQL

6	3	12	40+	15+
User Roles	Tech Layers	Core Modules	API Endpoints	DB Tables

Document Title	Workforce Management System — Full PRD
Prepared By	Salma Dariya — Intern / Project Trainee
Organization	CTDS
Frontend Stack	React 18 · Tailwind CSS · Shadcn/UI · React Router v6
Backend Stack	Supabase (Auth + Realtime + Storage) · PostgreSQL (DBMS)
Architecture	RESTful API + Supabase Row-Level Security (RLS) policies
User Roles	Super Admin · Admin · Operation Manager · HR · Dept Head · Employee
Version	1.0 — Final PRD
Date	June 2026
Status	Approved for Development

1. Executive Summary

The CTDS Workforce Management System (WMS) is a full-stack enterprise-grade web application designed to digitize, streamline, and secure every dimension of workforce operations — from employee lifecycle management and KPI tracking to task delegation and performance assessment. Built on React 18 with Tailwind CSS for the frontend, Supabase as the backend-as-a-service platform (Auth, Realtime, Storage), and PostgreSQL as the relational database engine, the system enforces a strict 6-tier role hierarchy secured by Supabase Row-Level Security (RLS) policies.

This PRD was researched and authored by Salma Dariya, Intern at CTDS, drawing from industry-standard workforce management frameworks (SHRM, ISO 9001 HR processes, ITIL service management), enterprise SaaS patterns (Workday, BambooHR, Jira), and modern web application security standards (OWASP, RBAC best practices).

Dimension	Specification
System Name	CTDS Workforce Management System (WMS)
Frontend	React 18 · Tailwind CSS v3 · Shadcn/UI · React Router v6 · Zustand (state)
Backend / BaaS	Supabase v2 (Auth · Database · Realtime · Storage · Edge Functions)
DBMS	PostgreSQL 15 with Row-Level Security (RLS) policies per role
Authentication	Supabase Auth — JWT tokens, email/password, session management
Authorization	6-tier RBAC enforced at DB level via RLS + frontend route guards
Deployment Target	Vercel (frontend) + Supabase Cloud (backend) + GitHub CI/CD
API Pattern	Supabase JS SDK (PostgREST auto-generated REST API)
Real-time	Supabase Realtime subscriptions for live task and KPI updates
File Storage	Supabase Storage buckets — employee documents, profile images

2. Problem Statement & Business Context

Modern organizations operating without an integrated workforce management platform face compounding inefficiencies that directly impact productivity, compliance, and employee satisfaction. CTDS currently operates with fragmented tools — spreadsheets for attendance, email chains for task delegation, and manual processes for KPI reviews — creating the following measurable pain points:

Pain Point	Business Impact	Industry Benchmark
No centralized employee database	Data inconsistency; 40% time lost to manual lookups	Industry: centralized HRIS reduces admin time by 35% (SHRM 2024)
Manual KPI creation and tracking	No accountability; missed performance cycles	Best-in-class: automated KPI dashboards reduce review cycles by 60%

Pain Point	Business Impact	Industry Benchmark
Task delegation via email/WhatsApp	No traceability; no deadline enforcement	Project management tools reduce task-related delays by 45% (PMI)
No role-based access control	Data privacy violations; compliance risk	GDPR/ISO 27001 requires strict data access segregation
No structured approval workflows	Bottlenecks in leave, promotions, onboarding	Workflow automation cuts approval times by 70% (McKinsey)
Paper/manual performance assessments	Bias, inconsistency, no audit trail	Digital assessment tools improve review quality by 55% (Gallup)
No real-time operational visibility for management	Leadership decisions based on stale data	Real-time dashboards improve decision speed by 3x (Deloitte)

3. System Architecture

3.1 Technology Stack — Detailed

Layer	Technology	Purpose	Version
UI Framework	React	Component-based SPA — dynamic role-specific dashboards	18.x
Styling	Tailwind CSS	Utility-first CSS — consistent design system, dark/light mode	v3.x
UI Components	Shadcn/UI	Accessible, headless components — tables, modals, forms	Latest
State Management	Zustand	Global auth state, role context, notification store	4.x
Routing	React Router	Protected role-based routes, lazy loading	v6
Forms	React Hook Form + Zod	Form validation, schema-based input validation	Latest
Charts	Recharts	KPI dashboards, performance charts, attendance trends	2.x
HTTP Client	Supabase JS SDK	Auto-generated REST API client, auth helpers	v2
Backend Platform	Supabase	Auth · PostgreSQL · Realtime · Storage · Edge Functions	v2
Database	PostgreSQL	Relational DBMS — RLS policies, triggers, functions	15.x
Authentication	Supabase Auth	JWT-based auth, session management, email verification	v2
File Storage	Supabase Storage	Employee documents, profile pictures, KPI attachments	v2
Real-time	Supabase Realtime	Live task updates, KPI notifications, online presence	v2
Deployment	Vercel + GitHub Actions	CI/CD pipeline, preview deployments, environment management	Latest

3.2 Architectural Pattern

The system follows a three-tier architecture: React SPA (Presentation Layer) → Supabase JS SDK (API Layer) → PostgreSQL with RLS (Data Layer). All business logic involving data security is enforced at the database level via Row-Level Security policies, ensuring that even if the frontend is compromised, unauthorized data access is impossible. Edge Functions handle complex server-side operations such as batch KPI calculations, email notifications, and payroll exports.

Tier	Components	Security Enforcement
Presentation	React 18 + Tailwind CSS + Zustand — role-specific UI rendering, route guards	Frontend route guards check JWT role claims; unauthorized routes redirect to 403
API	Supabase PostgREST — auto-generated REST endpoints from DB schema	Supabase Auth middleware validates JWT on every request; anonymous access blocked
Data	PostgreSQL 15 + RLS policies — per-table, per-role row filtering	RLS policies enforce data isolation at DB level — no bypass possible
Functions	Supabase Edge Functions (Deno) — complex business logic, notifications	Service role key — never exposed to client; internal-only execution
Storage	Supabase Storage — buckets with per-role access policies	Bucket policies mirror DB RLS — employees only access their own files

4. Database Design (PostgreSQL)

Complete relational schema with 15 core tables. All tables include created_at, updated_at (auto-managed by triggers), and soft-delete via is_deleted boolean. UUID primary keys throughout for security.

4.1 Core Tables

Table	Primary Key	Key Fields	RLS Policy Summary
profiles	uuid (FK → auth.users)	full_name, email, role, department_id, designation, phone, avatar_url, status, employee_id	SELECT: own row OR manager of dept OR admin+. INSERT/UPDATE/DELETE: Super Admin + Admin only
departments	uuid	name, head_id (FK→profiles), description, created_by	SELECT: all authenticated. CRUD: Super Admin, Admin, Dept Head (own dept)
roles	uuid	name (enum), permissions (jsonb), description	SELECT: all authenticated. CRUD: Super Admin only
tasks	uuid	title, description, assigned_to (FK→profiles), assigned_by, dept_id, priority, status, due_date, progress_pct, attachments[]	SELECT: own tasks OR dept head OR admin+. INSERT: Admin, Op Manager, Dept Head.

Table	Primary Key	Key Fields	RLS Policy Summary
			UPDATE: assignee (status/progress) + creator + admin+
kpis	uuid	title, description, target_value, unit, dept_id, assigned_to, created_by, period_start, period_end, weight	SELECT: own KPIs OR HR OR admin+. INSERT/UPDATE/DELETE: HR Manager, Super Admin, Admin
kpi_assessments	uuid	kpi_id, assessed_by, score, remarks, assessment_date, evidence_url	SELECT: own OR HR OR admin+. INSERT: HR, Dept Head. UPDATE/DELETE: HR, Admin+
attendance	uuid	employee_id, date, check_in, check_out, status, location_lat, location_lng, selfie_url, approved_by	SELECT: own OR dept head OR admin+. INSERT: employee (own). UPDATE: Admin, Op Manager
leave_requests	uuid	employee_id, leave_type, start_date, end_date, reason, status, approved_by, approved_at	SELECT: own OR manager. INSERT: employee. UPDATE (approve/reject): Op Manager, Admin+
announcements	uuid	title, content, target_roles (array), created_by, expires_at, priority	SELECT: roles match target_roles. CRUD: Admin+, Op Manager
notifications	uuid	recipient_id, type, title, message, is_read, metadata (jsonb), created_at	SELECT: own only. INSERT: system/edge functions. UPDATE (mark read): own only
documents	uuid	employee_id, type, file_url, file_name, uploaded_by, verified, verified_by	SELECT: own OR admin+. INSERT: employee (own), admin. UPDATE/DELETE: Admin+
payroll	uuid	employee_id, period, base_salary, allowances, deductions, net_pay, status, processed_by	SELECT: own (limited) OR HR OR admin+. CRUD: Super Admin, Admin only
audit_logs	uuid	actor_id, action, table_name, record_id, old_values (jsonb), new_values (jsonb), ip_address	SELECT: Super Admin only. INSERT: DB triggers (automatic). No UPDATE/DELETE
system_settings	uuid	key, value (jsonb), description, updated_by	SELECT: Admin+. UPDATE: Super Admin only
departments_kpi_targets	uuid	dept_id, kpi_id, target, period, set_by	SELECT: dept head (own) OR HR OR admin+. CRUD: HR, Admin+

4.2 Key PostgreSQL Functions & Triggers

Name	Type	Trigger / Usage
update_updated_at()	Trigger Function	Fires BEFORE UPDATE on all tables — auto-sets updated_at = NOW()
log_data_changes()	Trigger Function	Fires AFTER INSERT/UPDATE/DELETE on sensitive tables — writes to audit_logs
calculate_attendance_pct()	DB Function	Called by Edge Function — computes SUM attendance % per employee per period
get_employee_kpi_score()	DB Function	Aggregates weighted KPI scores into composite performance score
notify_on_task_assign()	Trigger Function	Fires AFTER INSERT on tasks — creates notification record for assignee
enforce_role_hierarchy()	DB Function	Validates that inserting admin cannot set a role higher than their own

5. Role Hierarchy & Permission Matrix

The system enforces a strict 6-tier role hierarchy. Higher roles inherit all permissions of lower roles unless explicitly restricted. Role assignment is constrained: no user can assign a role equal to or higher than their own.

Tier 1 — Super Admin

System owner. Unrestricted access to all data, settings, and audit logs. Cannot be created by any other role — provisioned directly in Supabase Auth.

Permissions: Full CRUD on all tables · Manage all users (create/read/update/delete) · Add/remove Admins · View audit logs · Modify system settings · Access payroll · Export all data · Impersonate users for support

Tier 2 — Admin

Organizational administrator. Manages day-to-day operations across all departments. Created by Super Admin only.

Permissions: CRUD on all employees (except Super Admin records) · Register Operation Managers, HR Managers, Dept Heads, Employees · Manage departments · Approve attendance & leave · View all KPIs and assessments · Access all dashboards · Cannot modify system settings or view raw audit logs

Tier 3 — Operation Manager

Oversees operational workflow. Manages tasks, approvals, and cross-department coordination. Registered by Admin.

Permissions: CRUD on tasks (create, assign, edit, delete) for all departments · Approve/reject leave requests · Register employees within assigned departments · View attendance across departments · View KPI summaries · Manage announcements · Cannot modify KPIs, payroll, or admin-level data

Tier 4 — HR Manager

Human resources authority. Full lifecycle of KPIs and performance assessments. Registered by Admin.

Permissions: Add, edit, remove KPIs (for any employee or department) · Conduct and record KPI assessments · View all employee profiles · Manage documents and contracts · View attendance and leave data · Generate performance reports · Cannot assign tasks, approve operational requests, or access payroll

Tier 5 — Department Head

Leads a specific department. Manages department employees and their tasks and performance. Registered by Admin.

Permissions: CRUD on employees within own department only · Create and assign tasks to department members · View KPIs and assessments for own department · Approve attendance for department members · View department-level performance dashboards · Cannot access other departments' data

Tier 6 — Employee

End user. Interacts with their own tasks, KPIs, and attendance. Registered by Admin, Op Manager, or Dept Head.

Permissions: View own assigned tasks · Update task status and progress percentage · View own KPI targets and scores · Submit attendance check-in/out · Apply for leave · View own documents · View announcements targeted to Employee role · Cannot view peer data

5.1 Permission Matrix

Module	Super Admin	Admin	Op Manager	HR Manager	Dept Head	Employee
Manage Super Admins	CRUD	—	—	—	—	—
Manage Admins	CRUD	Read	—	—	—	—
Manage Op Managers	CRUD	CRUD	—	—	—	—
Manage HR Managers	CRUD	CRUD	—	—	—	—
Manage Dept Heads	CRUD	CRUD	—	—	—	—
Manage Employees	CRUD	CRUD	CRUD (dept)	Read	CRUD (own dept)	—
Task Management	CRUD	CRUD	CRUD	Read	CRUD (dept)	Read+Update
KPI Management	CRUD	CRUD	Read	CRUD	Read (dept)	Read (own)
KPI Assessments	CRUD	CRUD	—	CRUD	Read+Add	Read (own)
Attendance	CRUD	CRUD	Approve	Read	Approve (dept)	Own only
Leave Requests	CRUD	CRUD	Approve	Read	Approve (dept)	Apply (own)
Payroll	CRUD	CRUD	—	Read (limited)	—	Read (own, limited)
Departments	CRUD	CRUD	Read	Read	Read (own)	Read
Announcements	CRUD	CRUD	CRUD	Read	Read	Read (targeted)

Module	Super Admin	Admin	Op Manager	HR Manager	Dept Head	Employee
Audit Logs	Read	–	–	–	–	–
System Settings	CRUD	Read	–	–	–	–
Reports & Exports	All	All departments	Operational	HR & Performance	Own department	–

6. Functional Modules — Detailed Specification

6.1 Authentication & Session Management

Built on Supabase Auth with JWT tokens. On login, the JWT payload includes the user's role claim (set via a Supabase Auth hook that reads from the profiles table). The React frontend reads this claim to render role-specific layouts and enforce client-side route guards. All role checks are duplicated server-side via RLS.

Feature	Specification
Login	Email + password via Supabase Auth. JWT issued with role claim in metadata.
Session	Auto-refresh tokens. 24-hour session. Idle timeout after 30 minutes.
Password Reset	Email magic link via Supabase. Enforced on first login for all user accounts.
Route Guards	React Router protected routes check Zustand auth store. Unauthorized access → /unauthorized.
Multi-tab Sync	Supabase Realtime auth listener syncs logout across browser tabs instantly.
Audit on Login	Every successful and failed login is recorded in audit_logs via DB trigger.
Super Admin Provision	First Super Admin provisioned via Supabase Dashboard — not via app UI.

6.2 User Management Module

Action	Who Can Perform	Details
Create Super Admin	Supabase Dashboard / Super Admin	Direct DB insert with role='super_admin'
Create Admin	Super Admin	UI form: name, email, phone, designation. Auto-sends email invite via Supabase Auth.
Create Operation Manager	Super Admin, Admin	UI form: name, email, dept assignment, designation. Email invite sent.
Create HR Manager	Super Admin, Admin	UI form: name, email, designation. HR module access auto-granted.
Create Dept Head	Super Admin, Admin	UI form: name, email, department (FK). Department head_id updated.
Create Employee	Super Admin, Admin, Op Mgr, Dept Head	Full form: personal info, employment info, department, designation, employee_id, documents.

Action	Who Can Perform	Details
Edit User	Own profile (limited) + Admin+	Admin+: all fields. Employee: phone, avatar only. Email changes require Super Admin.
Deactivate / Delete	Admin+ (deactivate). Super Admin (hard delete).	Soft delete via is_deleted=true. Hard delete removes auth.users record.
View User List	Per role — see Permission Matrix	Paginated table with search, filter by dept/role/status. Export to CSV: Admin+.

6.3 Task Management Module

Tasks follow a structured lifecycle: Draft → Assigned → In Progress → Under Review → Completed / Rejected. Supabase Realtime pushes status changes live to all stakeholders.

Field	Type	Validation Rule
title	varchar(200)	Required. Min 5 chars.
description	text	Required. Supports markdown rendering on frontend.
assigned_to	uuid FK	Must be active employee or dept head. Validated via DB constraint.
priority	enum	LOW MEDIUM HIGH CRITICAL. Default: MEDIUM.
status	enum	DRAFT → ASSIGNED → IN_PROGRESS → UNDER_REVIEW → COMPLETED REJECTED
due_date	date	Must be >= today on creation. Overdue tasks auto-flagged by cron Edge Function.
progress_pct	integer 0–100	Employee-editable. Triggers notification to assigner at 50% and 100%.
attachments	text[]	Supabase Storage URLs. Max 10 files, 10MB each.
dept_id	uuid FK	Auto-set from assigned_to employee's department.

6.4 KPI Management Module

The KPI module is the performance backbone of the system. KPIs are created by HR Managers, assigned to employees or departments, and assessed on a regular cycle. The system supports weighted composite scoring aligned with OKR (Objectives and Key Results) methodology.

Feature	Owner	Description
Create KPI	HR Manager, Admin, Super Admin	Define: title, description, target_value, unit (% , number, currency), weight (1–5), period, assigned_to or dept_id.
Edit KPI	HR Manager, Admin, Super Admin	Full edit. If assessments already exist, target changes require justification (recorded in audit_log).
Delete KPI	HR Manager, Admin, Super Admin	Soft delete only if no assessments. Hard delete: Super Admin only. Cascades to kpi_assessments.

Feature	Owner	Description
View KPIs	All roles (per RLS)	Employee: own KPIs only. Dept Head: dept KPIs. HR: all. Super Admin/Admin: all.
Assess KPI	HR Manager, Dept Head	Score entry (0 to target_value), remarks, evidence_url (uploaded file). Date auto-set.
View Assessments	HR, Admin, Super Admin, own Employee	Filterable by period, department, employee, score range.
KPI Dashboard	HR, Admin, Super Admin	Recharts bar/line charts: avg score by dept, top/bottom performers, trend over time.
Employee KPI Card	Employee, Dept Head, HR, Admin	Shows each KPI: target, current score, % achieved, trend indicator.

6.5 Attendance Module

Feature	Specification
Check-In	Employee submits check-in: timestamp auto-captured. Optional: GPS coords stored in location_lat/lng. Selfie upload to Supabase Storage.
Check-Out	Employee submits check-out. Duration auto-calculated. Status auto-set: Present / Late / Half-Day.
Admin Override	Admin / Op Manager can edit any attendance record with mandatory remark saved in audit_log.
Late Threshold	Configurable in system_settings (default: 09:15). Edge Function runs at midnight to auto-flag.
Attendance %	Calculated by get_attendance_pct() DB function. Used in Employee Performance Chart and KPI inputs.
Bulk Import	Admin can bulk import attendance via CSV. Validated against active employee list.
Leave Integration	Approved leave requests auto-create attendance records with status = ON_LEAVE.
Reports	Admin/HR: monthly summary by dept. Export PDF/CSV via Edge Function.

6.6 Leave Management Module

Feature	Specification
Apply for Leave	Employee submits: leave_type (Annual/Sick/Casual/Unpaid), start_date, end_date, reason. Notification sent to approver.
Approve / Reject	Op Manager or Dept Head (within own dept) or Admin. Status → APPROVED/REJECTED with remarks. Realtime notification to employee.
Leave Balance	Tracked in profiles.leave_balance (jsonb: {annual: n, sick: n, casual: n}). Auto-decremented on approval via DB trigger.
Leave Calendar	Admin/Op Manager: org-wide leave calendar view. Dept Head: dept view. Color-coded by leave type.
Policy Config	Super Admin sets annual leave entitlements per designation in system_settings.

6.7 Dashboard Module — Per Role

Role	Dashboard Contents
Super Admin	System health metrics · User counts by role · Recent audit log entries · Dept performance heatmap · Storage usage · Active sessions count
Admin	Org-wide headcount · Pending approvals (leave, attendance) · Task completion rate · Top/bottom KPI performers · Department summary cards · Recent joins/exits
Operation Manager	Active tasks board (Kanban by status) · Overdue tasks alert · Leave requests pending · Attendance rate today · Department announcements
HR Manager	KPI assessment due this week · Performance distribution chart (Recharts) · Employees without KPIs · Recent assessments · Leave trend chart · New hires this month
Department Head	Own department task board · Department attendance % · Team KPI summary · My tasks assigned · Pending leave requests for team · Announcements
Employee	My tasks (by status) · My KPI scores · My attendance this month · Leave balance · Recent announcements · Profile completion %

7. Supabase Row-Level Security (RLS) Design

Every table has RLS enabled. Policies use `auth.uid()` and a helper function `get_my_role()` that reads the current user's role from the profiles table. This ensures all access rules are enforced at the PostgreSQL level, not just in application code.

Policy Pattern	SQL Approach	Applied To
Own data only	<code>auth.uid() = employee_id OR auth.uid() = id</code>	attendance, leave_requests, notifications, documents
Own dept only	<code>dept_id IN (SELECT dept_id FROM profiles WHERE id = auth.uid())</code>	tasks (dept head), kpi_assessments (dept head)
Role-based SELECT	<code>get_my_role() IN ('admin','super_admin','hr_manager')</code>	payroll, kpis, all_employee_profiles
Hierarchy enforcement	<code>get_my_role_rank() > get_target_role_rank(new.role)</code>	profiles INSERT/UPDATE (role field)
Super Admin bypass	<code>get_my_role() = 'super_admin'</code>	audit_logs, system_settings, hard deletes
Realtime filters	Broadcast policies mirror SELECT policies	tasks, notifications, attendance

Helper Function: `get_my_role()` — SECURITY DEFINER function that returns the role of `auth.uid()` from the profiles table. Cached per transaction for performance. Used in all RLS policies.

8. API Endpoints (Supabase PostgREST + Edge Functions)

Supabase auto-generates REST endpoints for all tables via PostgREST. Custom business logic is handled via Edge Functions (Deno runtime). All endpoints require Authorization: Bearer <jwt> header.

Endpoint	Method	Auth Required	Description
/rest/v1/profiles	GET/POST/PATCH/DELETE	All roles (RLS)	User CRUD — RLS enforces visible rows per role
/rest/v1/tasks	GET/POST/PATCH/DELETE	All roles (RLS)	Task CRUD — employees see own; dept head sees dept
/rest/v1/kpis	GET/POST/PATCH/DELETE	All roles (RLS)	KPI management — HR full; employees read own
/rest/v1/kpi_assessments	GET/POST/PATCH/DELETE	All roles (RLS)	Assessments — HR/DeptHead write; Employee read own
/rest/v1/attendance	GET/POST/PATCH	All roles (RLS)	Attendance log — employee inserts own; admin patches
/rest/v1/leave_requests	GET/POST/PATCH	All roles (RLS)	Leave management — approval via PATCH status field
/rest/v1/announcements	GET/POST/PATCH/DELETE	All roles (RLS)	Filtered by target_roles array match
/rest/v1/notifications	GET/PATCH	Own user only	Read notifications; PATCH marks is_read=true
/rest/v1/audit_logs	GET	Super Admin only	Read-only audit trail — no write via API
/functions/v1/create-user	POST	Admin+	Edge Function: creates Supabase Auth user + profiles row atomically
/functions/v1/calculate-performance	POST	HR, Admin+	Edge Function: aggregates KPI scores + attendance → performance report
/functions/v1/send-notification	POST	Internal only	Edge Function: triggered by DB — sends email/in-app notification
/functions/v1/export-report	POST	Admin+, HR	Edge Function: generates PDF/CSV report and uploads to Storage
/functions/v1/bulk-import	POST	Admin+	Edge Function: validates and imports CSV data for attendance/employees

9. Frontend Architecture (React + Tailwind CSS)

9.1 Route Structure

Route	Component	Access
/login	LoginPage	Public — redirects to /dashboard if authenticated
/dashboard	RoleDashboard (lazy)	All authenticated users — renders role-specific dashboard component
/users	UsersPage	Super Admin, Admin only
/users/new	CreateUserForm	Super Admin, Admin, Op Manager, Dept Head (restricted form fields)
/users/:id	UserProfile	Own profile: all. Others: Admin+, Dept Head (own dept)
/tasks	TasksPage	All roles — filtered by RLS
/tasks/new	CreateTaskForm	Admin+, Op Manager, Dept Head
/tasks/:id	TaskDetail	Assignee + assigner + Admin+
/kpis	KPIsPage	HR, Admin+, Dept Head (read)
/kpis/new	CreateKPIForm	HR, Admin+
/kpis/:id/assess	AssessKPIForm	HR, Dept Head (own dept)
/attendance	AttendancePage	All roles — filtered by role
/attendance/checkin	CheckInPage	Employee only
/leave	LeavePage	All roles
/departments	DepartmentsPage	Admin+
/reports	ReportsPage	HR, Admin+
/settings	SettingsPage	Super Admin only
/audit-logs	AuditLogsPage	Super Admin only
/unauthorized	UnauthorizedPage	Public — shown on role mismatch

9.2 Component Architecture

Component Category	Components	Notes
Layout	AppShell, Sidebar, TopNav, BreadcrumbBar, PageHeader	Role-aware sidebar renders only permitted nav items. Collapsible on mobile.
Auth	LoginForm, ProtectedRoute, RoleGuard, SessionProvider	Zustand auth store. JWT decoded on mount. Role stored in context.
Data Tables	DataTable (Shadcn), Pagination, FilterBar, ExportButton	Virtualized rows for large datasets. Column visibility toggles per role.
Forms	UserForm, TaskForm, KPIForm, AssessmentForm, AttendanceForm	React Hook Form + Zod schemas. Field visibility controlled by user's role.

Component Category	Components	Notes
Charts	KPIDashboard, PerformanceChart, AttendanceTrend, TaskProgressBar	Recharts. Responsive. Dark/light mode support via Tailwind CSS variables.
Modals	ConfirmModal, ViewModal, CreateModal, AssessModal	Shadcn Dialog. Accessible. Keyboard navigable.
Notifications	NotificationBell, NotificationDrawer, ToastProvider	Realtime subscription. Unread badge. Grouped by type.
KPI Cards	KPICard, KPIProgressBar, AssessmentHistory, ScoreBadge	Color-coded by achievement %. Green ≥80%, Amber 50–79%, Red <50%.

10. Non-Functional Requirements

Category	Requirement	Target Metric
Performance	Page load time (LCP)	< 2.5 seconds on 4G connection
Performance	API response time (P95)	< 300ms for all PostgREST queries
Scalability	Concurrent users	500+ concurrent sessions without degradation (Supabase Pro tier)
Security	Authentication	JWT with 24h expiry + refresh tokens. No plain-text passwords.
Security	Data isolation	PostgreSQL RLS — zero cross-role data leakage. Pen-test required pre-launch.
Security	OWASP Top 10	All critical vulnerabilities addressed. Input sanitization via Zod.
Reliability	Uptime SLA	99.9% uptime (Supabase Cloud SLA)
Reliability	Backups	Daily automated PostgreSQL backups via Supabase. Point-in-time recovery.
Accessibility	WCAG compliance	WCAG 2.1 AA — Shadcn/UI components are accessible by default
Responsiveness	Mobile support	Fully responsive via Tailwind CSS. Tested: 375px–1920px viewports.
Maintainability	Code quality	ESLint + Prettier. TypeScript strict mode. Test coverage ≥ 70%.
Compliance	Data privacy	GDPR-aligned: consent, right to access, right to erasure via soft delete.
Auditability	All data changes logged	audit_logs table auto-populated via DB triggers. Immutable via RLS.

11. Development Milestones & Sprint Plan

Sprint	Duration	Deliverables	Owner
Sprint 0 — Setup	Week 1	Supabase project init · PostgreSQL schema creation · RLS policies · GitHub repo · CI/CD pipeline · Tailwind design system tokens	Tech Lead
Sprint 1 — Auth & Users	Week 2	Login page · JWT role routing · Super Admin + Admin user management screens · Protected routes · Notification system base	Frontend + Backend
Sprint 2 — Core Roles	Week 3	Op Manager, HR Manager, Dept Head, Employee registration flows · Role dashboards (all 6) · Profile management	Frontend
Sprint 3 — Tasks	Week 4	Task CRUD (all roles) · Kanban board · Realtime status updates · Task detail page · Overdue detection Edge Function	Full Stack
Sprint 4 — KPI Module	Week 5	KPI creation/edit/delete (HR) · KPI assignment · Assessment form · KPI dashboard charts · Employee KPI card	Full Stack
Sprint 5 — Attendance	Week 6	Check-in/out · GPS + selfie capture · Leave management · Attendance reports · Leave calendar	Full Stack
Sprint 6 — Reports	Week 7	Performance report generation (PDF/CSV) · Audit log viewer · System settings · Bulk import · Email notifications	Backend + Frontend
Sprint 7 — QA & Launch	Week 8	End-to-end testing · Security pen-test (RLS audit) · Performance optimization · User acceptance testing · Production deployment	QA + DevOps

12. Sign-Off & Approval

Role	Name	Signature	Date
Intern / Author	Salma Dariya		June 2026
Project Supervisor			
Technical Lead			
Senior Management			

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